

Liquid Extraction

Final Lab Report

Unit Operations Lab II

February 19, 2020

Group 3, Thursday, AM section

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1. Executive Summary

The objective of this experiment is to separate a 0.5 wt% mixture of acetic acid and white oil using water by operating the three-stage mixer settler. Samples collected from each stage will be used to investigate how the speed of mixing effects the efficiency of extraction.

First, white oil contaminated with acetic acid was pumped through the liquid extraction using the lowest agitator speed of 400 RPM. Water flowed counter-current to the white oil through three stages. After 15 minutes, samples of water and oil were taken for each stage. These samples were covered with parafilm and titrated with NaOH to determine the concentration of acetic acid in the sample.

Using the data collected, plots were created of the concentration of acetic acid versus stage number and agitator speed. Next, a plot of acetic acid in the extract versus the acetic acid in the raffinate stage for each agitator speed was created. Next, a McCabe-Thiele plot was created to obtain the number of ideal stages. Next, the Murphree efficiency for each stage and the overall efficiency was calculated.

The results showed that with increasing stage number, the concentration of acetic acid decreased for the extract and the raffinate. Additionally, as the agitator speed increases, there is more mass transfer of acetic acid from the raffinate to the extract. From the McCabe-Thiele plot, the theoretical number of plates is 1. Next, the Murphree efficiency decreased as each stage increased. As agitator speed increased, the overall Murphree efficiency decreased.

In conclusion, our results sometimes agree with the theory, however, there is some experimental error. Against the theory, no agitator speed was more efficient than the other.

2. Introduction

In this lab, a three-stage mixer-settler is used to extract acetic acid from white oil by using water. The solution of 0.5wt% acetic acid in white oil is pumped from a storage tank through three stages. Each stage has an agitator and water that flows counter-current to the oil. As the white oil mixture passes through each stage, the acetic acid dissolves into the water while the white oil remains unaffected by water. This is because the two solvents (water and white oil) are immiscible. The white oil which has been stripped of acetic acid will leave the third stage as the raffinate and will be collected in another storage tank. Titrations of extract and raffinate samples collected at different stages should show different concentrations of acetic acid. The concentration of acetic acid separated will also be affected by the agitator speeds in the stage drums. There are three motorized agitators mounted on the top of each stage and has variable speed. The speed can be adjusted by dials attached to the agitators but must be measured by laser speedometer. [1]

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Along with distillation, liquid extraction is a very common method of separation used in a variety of industries, such as pharmaceutical, mineral extraction, and environmental areas. It's important to note that both liquid extraction and distillation should share similar analysis methods, since both types of separations should have phase equilibrium at each stage of the process.

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3. Theory

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The main principle behind liquid extractions is the mass transfer of a solute from a certain liquid, into a solvent that is miscible with the solute. The liquid that will experience extraction is defined as the feed, the liquid containing the extracted solute is defined as the extract, and the liquid which has had the solute removed is defined as the raffinate. For a staged system of liquid extraction, the concentration of solute in the feed for a particular stage, $n+1$ can be defined as:

$$y_{n+1} = \left(\frac{E^{n+1}-1}{E-1} \right) * y_1 \quad (1)$$

Where:

y_1 = initial concentration of solute in feed

$$E = \frac{KL}{H}$$

H = feed solvent

K = equilibrium constant

L = extraction solvent

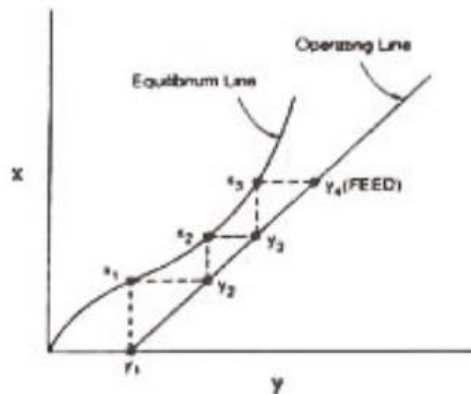


Figure: 1 [2]

(1) can be derived from a linear relationship of phase equilibrium with one stage [3]:

$$x = Ky \quad (2)$$

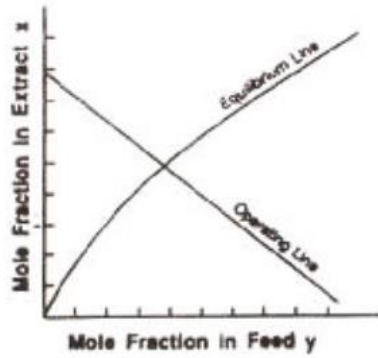


Figure: 2 [2]

Where:

x = product concentration in extraction solvent L

y = product concentration in feed solvent H

Since liquid extraction is analogous to distillation, the efficiency of separation can be defined as σ :

$$\sigma = \frac{N_A}{N_T} \quad (3)$$

Where:

N_A = Actual number of stages

N_T = Theoretical number of stages

For efficiencies of extract, raffinate and overall at various stages, the following relationships are used:

$$E_{ME} = \left(\frac{Y_{S+1} - Y_S}{Y_{S+1}^* - Y_S} \right) \quad (4)$$

Where:

E_{ME} = Murphree efficiency of extract

Y_{S+1}^* = composition of extract phase in equilibrium with raffinate phase leaving stage S+1

$$E_{MR} = \left(\frac{X_S - X_{S+1}}{X_S - X_{S+1}^*} \right) \quad (5)$$

Where:

E_{MR} = Murphree efficiency of raffinate

X_{S+1}^* = composition of raffinate phase in equilibrium with extract phase leaving stage S+1

$$E_0 = \frac{N_T}{N_A} \quad (6)$$

Where:

E_0 = Overall efficiency

N_T = Theoretical number of stages required

N_A = Actual number of stages used

It should be expected for acetic acid to follow a linear equilibrium composition in a raffinate vs extract phase plot given by the equation: $Y = 34.106 X$. [4]

An example of liquid extraction from industry is demonstrated by Koch Modular. This company uses Liquid-Liquid Extraction as an initial step to purify algae so it can be used to develop biofuels. This method is beneficial over distillation because in this case, an azeotrope would normally form. [5]

4. Results

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The goal of this lab is to understand the relationship of changing agitating rates and the separation of white oil and acetic acid. In the liquid-liquid separation unit, the extract (water) and raffinate (oil) streams are countercurrent. To determine the acetic acid concentration, and thus the separation of acetic acid from white oil, samples of both the extract and raffinate streams were taken at each stage. The samples were taken when steady state was reached in the mixer-

settler unit by waiting at least fifteen minutes after the agitator speeds were applied. The agitator speeds chosen were 400 RPM, 650 RPM, and 850 RPM.

At each stage for all trials (different agitator speeds), samples of 25 mL were taken of oil and water. The raw data collected for 400RPM, 650 RPM, and 850 RPM, and shown in **Tables 1, 2, and 3** respectively. From the samples, the mole fraction and wt% of acetic acid were calculated as shown in **Appendix III**. The calculation results are shown for each agitator speed respectively in **Tables 4, 5, and 6**.

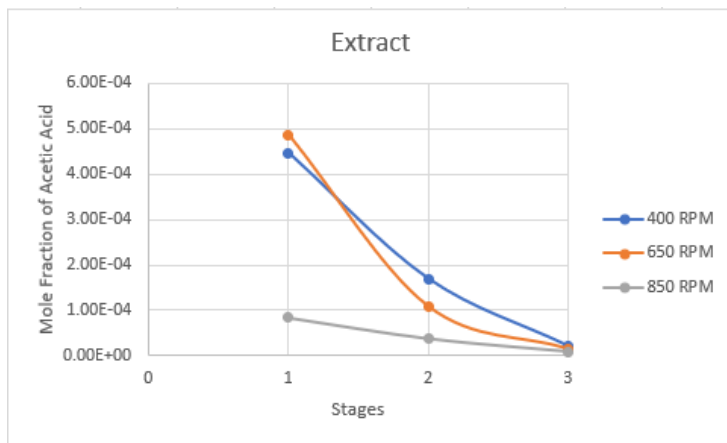


Figure 3: Mole Fraction of Acetic Acid in Extract at Different Agitator Speeds and Stages

Using these calculations, a relationship between the mole fraction of acetic acid and both stage number and agitator speed can be determined. **Figure 3** shows the relationship for the extract (water). The highest concentration of acetic acid is shown to be at the first stage of the mixer-settler unit using all agitator speeds. The concentration of acetic acid then decreases at each further stage, with stage 3 having the lowest concentration of acetic acid. As a function of agitator speed, as the agitator speed increases, the concentration of acetic acid in the extract at each stage should decrease as well. The data shows this direct relationship, relating it to theory. It can be noticed that the change in concentration of acetic acid from stage to stage at each agitator speed lessens as there are more stages. This is shown in **Figure 3** by the convergence of all acetic acid concentrations at stage 3. Thus, there is a decreasing rate of acetic acid at each further stage, converging at the mass transfer limitation.

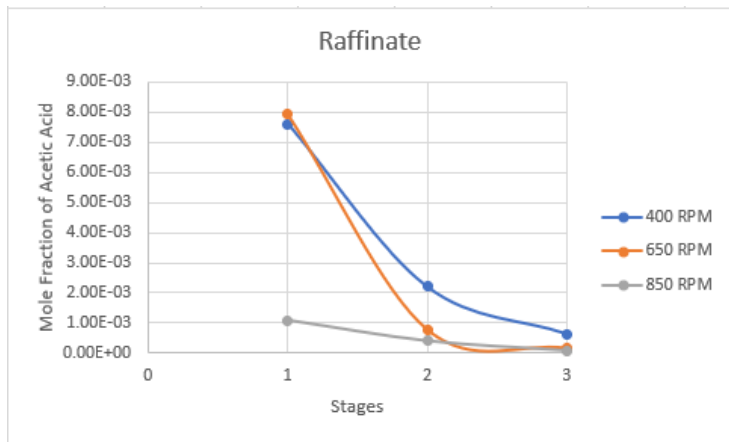


Figure 4: Mole Fraction of Acetic Acid in Raffinate at Different Agitator Speeds and Stages

Similarly, **Figure 4** shows the relationship for the raffinate (white oil). Using these calculations, a relationship between the mole fraction of acetic acid and both stage number and agitator speed can be determined. The highest concentration of acetic acid is shown to be at the first stage of the mixer-settler unit using all agitator speeds. The concentration of acetic acid then decreases at each further stage, with stage 3 having the lowest concentration of acetic acid. As a function of agitator speed, as the agitator speed increases, the concentration of acetic acid in the extract at each stage should decrease as well. The data shows this direct relationship, relating it to theory. It can be noticed that the change in concentration of acetic acid from stage to stage at each agitator speed lessens as there are more stages. This is shown in **Figure 4** by the convergence of all acetic acid concentrations at stage 3. Thus, there is a decreasing rate of acetic acid at each further stage, converging at the mass transfer limitation.

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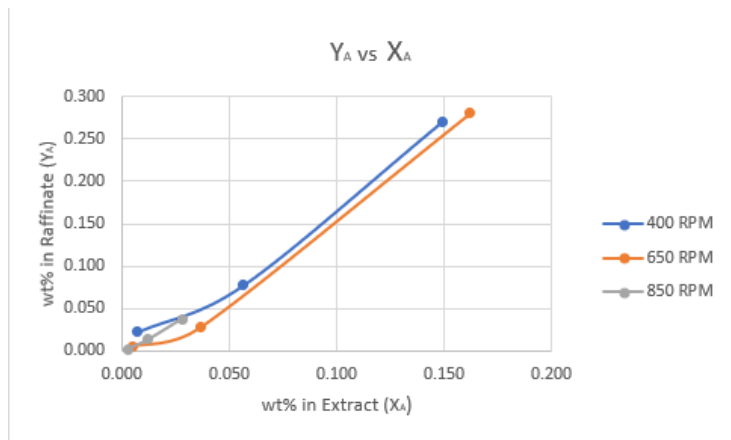


Figure 5: Y_A vs X_A of Acetic Acid at Different Agitator Speeds

Figure 5 shows the relationship between the mass transfer between the raffinate (Y) and the extract (X). As the acetic acid is transferred to the extract, the raffinate decreases in acetic acid mass. This shows the inverse relationship between the raffinate and extract. This is basically explaining the main concept of liquid-liquid extraction. When comparing the three different agitator speeds to the mass transfer, a lower agitator speed correlates to less mass transfer of acetic acid from the raffinate to the extract. As the agitator speed increases, there is more mass transfer of acetic acid from the raffinate to the extract. This can be shown by the ratio of Y_A and X_A in Figure 5. As the ratio approaches 1, the closer the limitation of mass transfer is reached.

The results were then analyzed using the McCabe-Thiele Diagram. This analysis will give the stage efficiency of the mixer-settler unit. Maximum efficiency is determined if the theoretical stages found from the McCabe-Thiele diagram yields the same number of actual stages used, three.

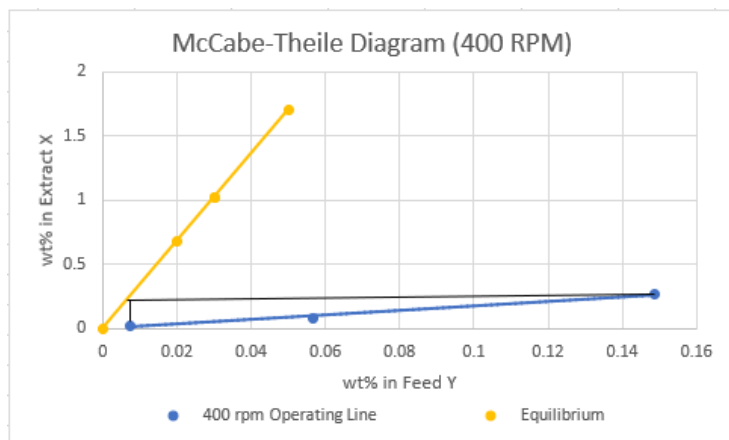


Figure 6: McCabe-Thiele Diagram for 400 RPM

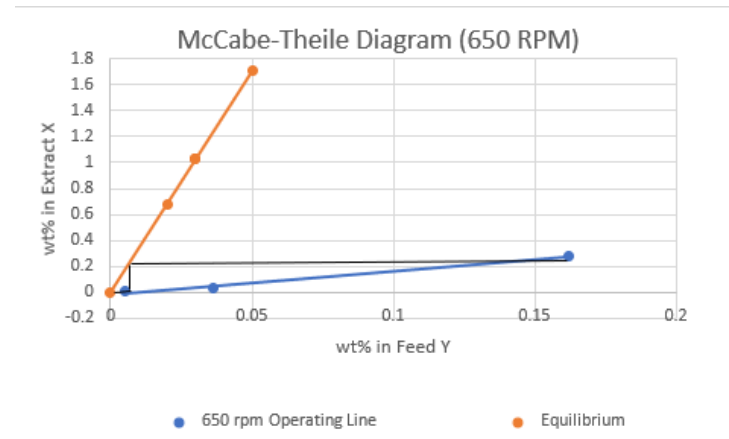


Figure 7: McCabe-Thiele Diagram for 650 RPM

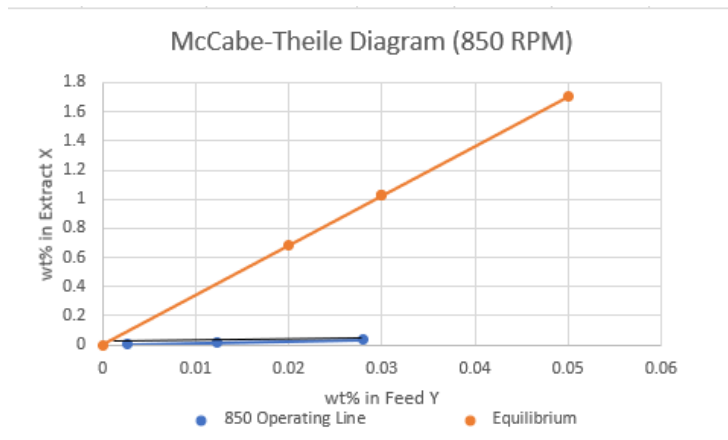


Figure 8: McCabe-Thiele Diagram for 850 RPM

Theoretical stages for each agitator speed trial was found and compared to the actual three stages used. **Figure 6** shows that there is one theoretical stage. **Figure 7** shows 1.07 theoretical stages for an agitator speed of 650 RPM. For 850 RPM, the theoretical stages in **Figure 8** shows 0.66 theoretical stages.

Thus, the McCabe-Thiele analysis did not reach a conclusion as the data collected is skewed. The only hypothesis that can be made from the results of the McCabe-Thiele diagram is from **Figure 6** and **Figure 7**. As the agitator speed increased, so did the number of theoretical stages. From using the results of theoretical stages given by 400 RPM and 650 RPM, this hypothesis correlated to theory of higher stage efficiency with higher agitator speeds.

To further analyze the efficiency of the stages in the liquid-liquid extraction, the Murphree efficiency at each stage and overall efficiency are calculated. The efficiencies are calculated using **Equation 4** for the extract and **Equation 5** for the raffinate and are shown to be calculated in **Appendix III**.

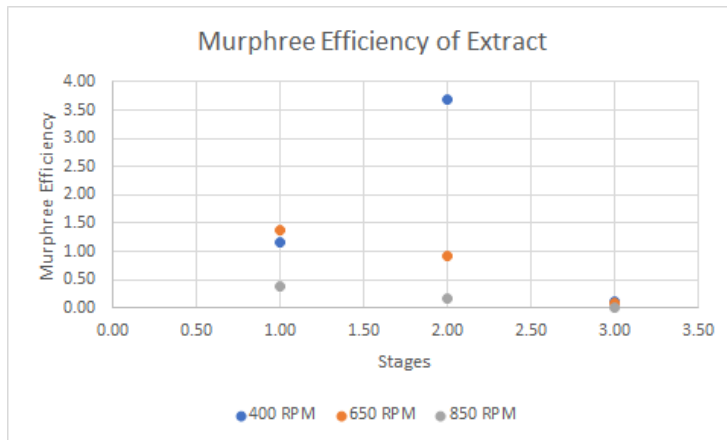


Figure 9: Murphree Efficiency of Extract

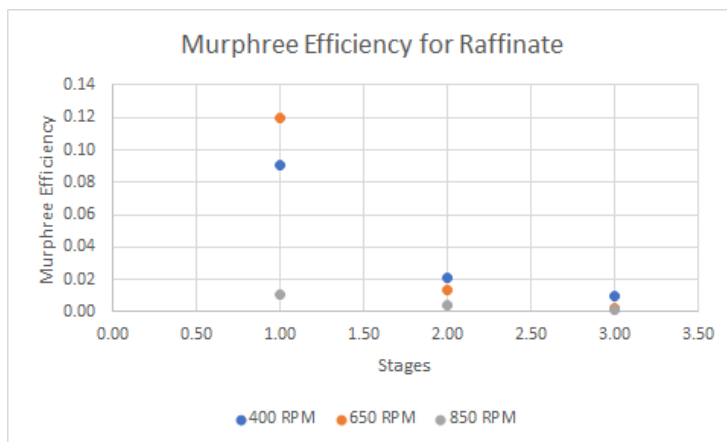


Figure 10: Murphree Efficiency for Raffinate

As seen in **Figure 9**, the Murphree efficiency of each stage is decreasing per each additional stage for the extract. As the agitator speed increases, the efficiency decreases per stage. In **Figure 10**, the Murphree efficiency of each stage also decreases per each additional stage for the raffinate, similar to the extract. As the agitator speed increases, the efficiency decreases per stage. Both for the extract and the raffinate, the Murphree efficiency seems to converge to zero as more stages are added.

The overall efficiencies were calculated using **Equation 6** using the theoretical plates from the McCabe-Thiele Diagrams. The results are shown below in Table 7. From agitator speeds 400

RPM to agitator speed of 650 RPM, the efficiency increases. However, when including 850 RPM, the efficiency decreases. Thus, no conclusion can be made on how agitator speed effects efficiency. Although, by theory, the agitator speed should increase the overall efficiency of the process, as more acetic acid is separated out of the feed.

Table 6: Overall Efficiency

Overall Efficiency	
Agitator Speed (RPM)	Overall Efficiency
400.00	0.33
650.00	0.36
850.00	0.22

5. Discussion

In this lab, the focus was on liquid-liquid extraction. The three-stage mixer-settler system was used to extract acetic acid from mineral oil while using water as the solvent [2]. This provided the relationship of how the mixing speed affects the efficiency of extracting the acetic acid from mineral oil. Three different mixing speeds were applied at 400, 650, 850 RPM. The process was run at each flowrate until it reaches equilibrium. At this point two samples were collected. One sample was the acetic acid in the water at the specific stage, and the other sample was the acetic acid in the mineral oil at the specific stage. Each sample was sent to be titrated against a standardized NaOH solution using a phenolphthalein indicator. This was repeated for all samples collected at each stage for every water flowrate applied.

The results show that as the number of stages and agitator speeds were increased, the mole fraction of acetic acid decreased in the raffinate and extract. Furthermore, the results show that as the acetic acid is transferred to the extract, the raffinate decreases in the mass of acetic acid. This shows the inverse relationship between the raffinate and extract. Next, as the agitator speeds increased, there is more mass transfer of acetic acid from the raffinate to the extract. Lastly, using the McCabe-Thiele Diagram to find the number of theoretical stages, the number obtained was 1 stage for 400 RPM and 1.07 stages for 650 RPM. However, due to possible error the 850 RPM flowrate had 0.66 stages. The actual number of stages we used was 3 stages. In conclusion, as the flowrate increases from 400 RPM to 650 RPM, the number of theoretical stages goes up and gives a higher efficiency until 850 RPM is applied. Lastly, looking at the Murphree efficiencies, the conclusion made is the efficiency of each stage is decreasing as the stages are going up in the extract and raffinate. Furthermore, as the agitator speeds are increasing the efficiency is decreasing.

Our results do agree with the theory on how the different flowrates of the solvent effect the extent of extraction. It agrees with the theory on how the number of stages and agitator speeds were increasing, the mole fraction of acetic acid is decreasing in the raffinate and extract. Furthermore, it agrees with the theory of higher stage efficiency with higher agitator speeds. Our results for the Murphree efficiency do not agree with the theory because increasing agitator

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how would your system change if reflux is introduced? will the efficiency decrease or increase?

speed should result in increasing efficiency. This disagreement is mostly likely due to an error in sample collection.

The most probable sources of error could be from two huge reasons. The first reason would be collecting the samples. Every sample was collected to be 25 mL. However, due to trouble with pipetting, this measurement was not always constant throughout the experiment. Therefore, this could have skewed the results after titrating. Furthermore, parafilm may not have been applied fast enough on the sample once collected, which could have led to some evaporation and lowered the concentration of acetic acid. Our results do not show a problem with the experiment plan. Due to the possible errors listen above, the 850 RPM flowrate gave an inaccurate number of stages that do not follow the trend. Therefore, one question that remains unanswered is what were the specific errors which led to this inaccuracy at the 850 RPM flowrate.

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6. References

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[1] ChE 382 Supplementary information: Solvent extraction Unit Diagram, Spring 2019, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.

[2] ChE 382: Unit Operations Laboratory – Liquid Extraction, Spring 2019, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.

[3] Wankat, Phillip C. "Separation Process Engineering : Includes Mass Transfer Analysis." International Journal of Environmental Studies, vol. 66, no. 6, 2012.

[4] ChE 382 Supplementary information: Liquid Extraction Water-White Oil-Acetic Acid Equilibrium Data, Spring 2019, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.

[5] Modular, Koch. *Typical Industrial Applications for Separation by Liquid-Liquid- Extraction*. <https://kochmodular.com/our-work/expertise/>. Accessed 19 Feb. 2020.

7. Appendix I: Data Tabulation/Graphs

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Table 1: Raw Data for 400 RPM

400		
Stage	Sample	NaOH (mL)
Feed	Initial	137.2
1		
Extract	1	61.8
Raffinate	1	95.7
2		
Extract	2	23.5
Raffinate	2	27.6
3		
Extract	3	3
Raffinate	3	8

Table 2: Raw Data for 650 RPM

650		
Stage	Sample	NaOH (mL)
1		
Extract	4	67.2
Raffinate	4	99.7
2		
Extract	5	15.1
Raffinate	5	9.8
3		
Extract	6	2.2
Raffinate	6	2.1

Table 3: Raw Data for 850 RPM

850		
Stage	Sample	NaOH (mL)
1		
Extract	4	11.6
Raffinate	4	13.5
2		
Extract	5	5.1
Raffinate	5	4.9
3		
Extract	6	1.1
Raffinate	6	0.9

Table 4: Calculations for 400 RPM

Agitator Speed [RPM]	400								
Stage	1	Extract	Raffinate	2	Extract	Raffinate	3	Extract	Raffinate
Sample Volume [mL]		25.00	25.00		25.00	25.00		25.00	25.00
NaOH Delivered [mL]		61.8	95.7		23.5	27.6		3	8
NaOH Delivered [L]		6.18E-02	9.57E-02		2.35E-02	2.76E-02		3.00E-03	8.00E-03
Moles of NaOH		6.18E-04	9.57E-04		2.35E-04	2.76E-04		3.00E-05	8.00E-05
Moles of Acetic Acid		6.18E-04	9.57E-04		2.35E-04	2.76E-04		3.00E-05	8.00E-05
Mass of Acetic Acid [g]		0.0371	0.0575		0.0141	0.0166		0.0018	0.0048
Mass of Water/Oil [g]		24.89	21.28		24.91	21.31		24.92	21.32
Moles of Water/Oil		1.382	0.125		1.383	0.125		1.383	0.125
Mole Fraction of Acetic Acid		4.47E-04	7.60E-03		1.70E-04	2.20E-03		2.17E-05	6.39E-04
Mass Fraction of Acetic Acid		1.49E-03	2.69E-03		5.66E-04	7.77E-04		7.23E-05	2.25E-04
wt% of Acetic Acid		0.149	0.269		0.057	0.078		0.007	0.023

Table 5: Calculations for 650 RPM

Agitator Speed [RPM]	650								
Stage	1	Extract	Raffinate	2	Extract	Raffinate	3	Extract	Raffinate
Sample Volume [mL]		25.00	25.00		25.00	25.00		25.00	25.00
NaOH Delivered [mL]		67.2	99.7		15.1	9.8		2.2	2.1
NaOH Delivered [L]		6.72E-02	9.97E-02		1.51E-02	9.80E-03		2.20E-03	2.10E-03
Moles of NaOH		6.72E-04	9.97E-04		1.51E-04	9.80E-05		2.20E-05	2.10E-05
Moles of Acetic Acid		6.72E-04	9.97E-04		1.51E-04	9.80E-05		2.20E-05	2.10E-05
Mass of Acetic Acid [g]		0.0404	0.0599		0.0091	0.0059		0.0013	0.0013
Mass of Water/Oil [g]		24.89	21.28		24.92	21.32		24.92	21.32
Moles of Water/Oil		1.381	0.125		1.383	0.125		1.383	0.125
Mole Fraction of Acetic Acid		4.86E-04	7.92E-03		1.09E-04	7.82E-04		1.59E-05	1.68E-04
Mass Fraction of Acetic Acid		1.62E-03	2.81E-03		3.64E-04	2.76E-04		5.30E-05	5.91E-05
wt% of Acetic Acid		0.162	0.281		0.036	0.028		0.005	0.006

Table 5: Calculations for 850 RPM

Agitator Speed [RPM]	850							
Stage	1 Extract	Raffinate	2 Extract	Raffinate	3 Extract	Raffinate		
Sample Volume [mL]	25.00	25.00	25.00	25.00	25.00	25.00		
NaOH Delivered [mL]	11.6	13.5	5.1	4.9	1.1	0.9		
NaOH Delivered [L]	1.16E-02	1.35E-02	5.10E-03	4.90E-03	1.10E-03	9.00E-04		
Moles of NaOH	1.16E-04	1.35E-04	5.10E-05	4.90E-05	1.10E-05	9.00E-06		
Moles of Acetic Acid	1.16E-04	1.35E-04	5.10E-05	4.90E-05	1.10E-05	9.00E-06		
Mass of Acetic Acid [g]	0.0070	0.0081	0.0031	0.0029	0.0007	0.0005		
Mass of Water/Oil [g]	24.92	21.32	24.92	21.32	24.92	21.32		
Moles of Water/Oil	1.383	0.125	1.383	0.125	1.384	0.125		
Mole Fraction of Acetic Acid	8.39E-05	1.08E-03	3.69E-05	3.91E-04	7.95E-06	7.19E-05		
Mass Fraction of Acetic Acid	2.79E-04	3.80E-04	1.23E-04	1.38E-04	2.65E-05	2.53E-05		
wt% of Acetic Acid	0.028	0.038	0.012	0.014	0.003	0.003		

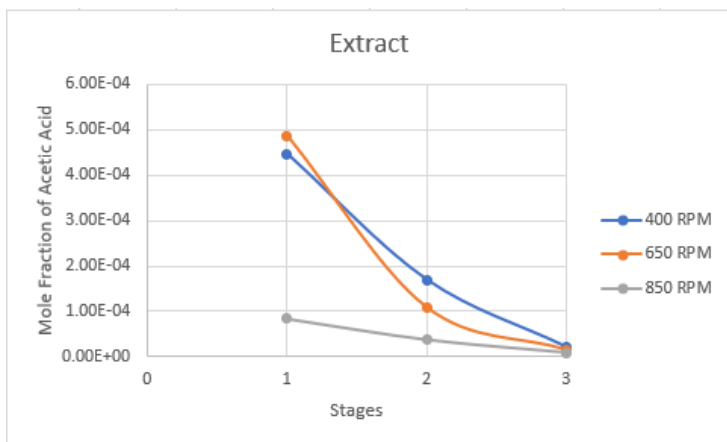


Figure 3: Mole Fraction of Acetic Acid in Extract at Different Agitator Speeds and Stages

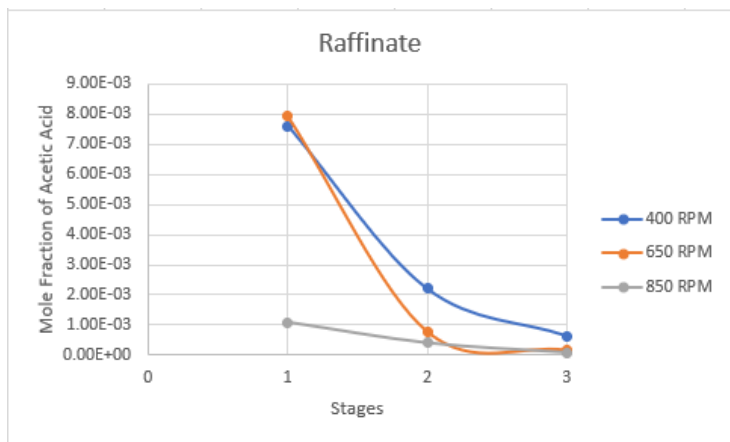


Figure 4: Mole Fraction of Acetic Acid in Raffinate at Different Agitator Speeds and Stages

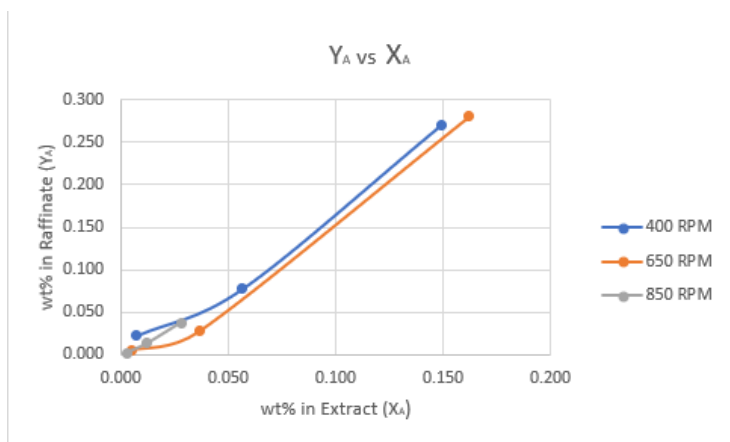


Figure 5: Y_A vs X_A of Acetic Acid at Different Agitator Speeds

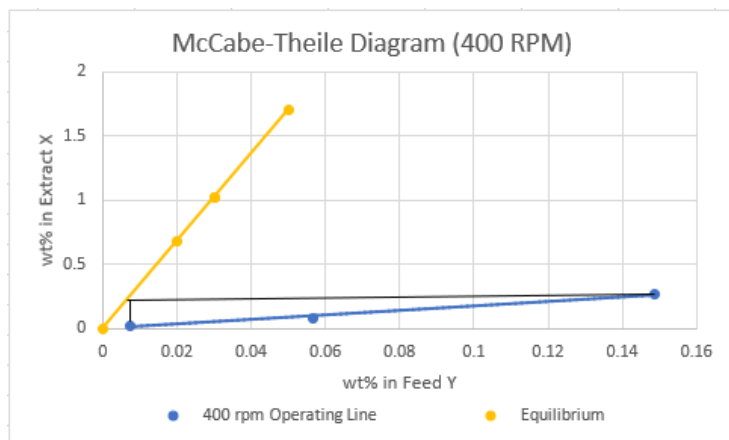


Figure 6: McCabe-Thiele Diagram for 400 RPM

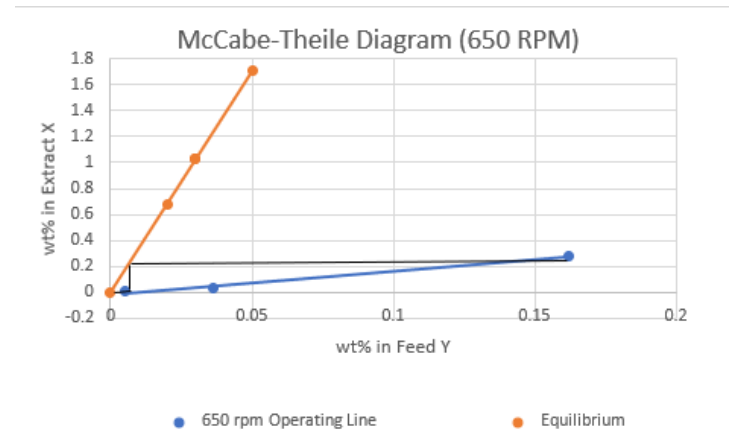


Figure 7: McCabe-Thiele Diagram for 650 RPM

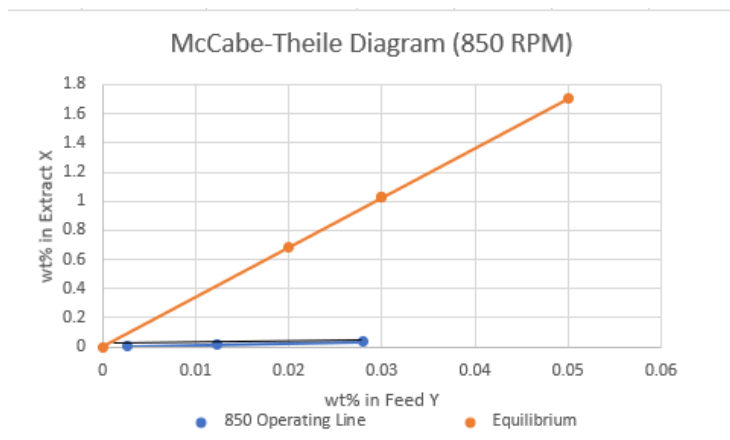


Figure 8: McCabe-Thiele Diagram for 850 RPM

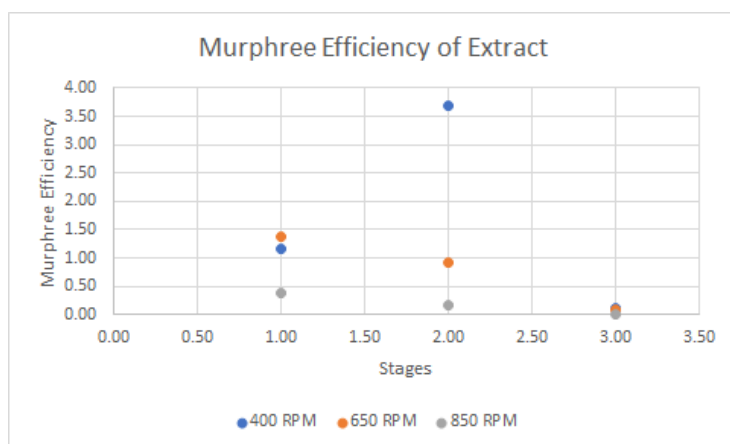


Figure 9: Murphree Efficiency of Extract

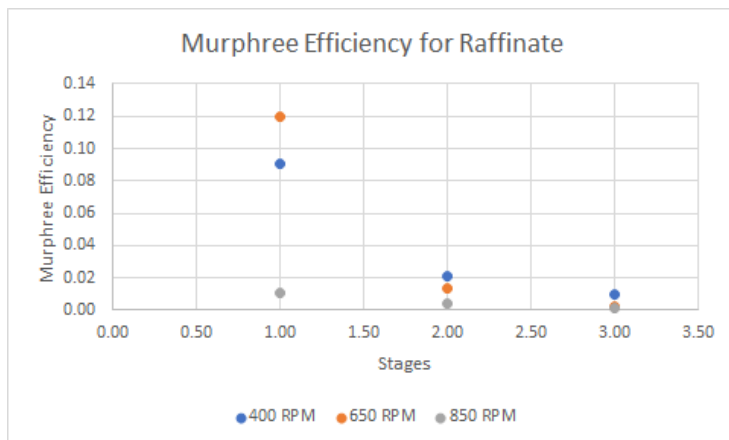


Figure 10: Murphree Efficiency for Raffinate

Table 6: Overall Efficiency

Overall Efficiency	
Agitator Speed (RPM)	Overall Efficiency
400.00	0.33
650.00	0.36
850.00	0.22

8. Appendix II: Error Analysis

The most probable sources of error could be from two huge reasons. The first reason would be collecting the samples. Every sample was collected to be 25 mL. However, due to trouble with pipetting, this measurement was not always constant throughout the experiment. Therefore, this could have skewed the results after titrating. Furthermore, parafilm may not have been applied fast enough on the sample once collected, which could have led to some evaporation and messed with the results.

Commented [PTH14]: 7/10 More quantitative analysis needed here. What about the error in the titrations? Error in the oil and water flowrates?

9. Appendix III: Sample Calculations

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The amount of Acetic acid added initially Givens:

$$1 \text{ US gallon} = 3785.41 \text{ cm}^3$$

$$\text{White Oil Density} = 0.8 \frac{\text{g}}{\text{cm}^3}$$

$$\text{Acetic Acid Density} = 1.05 \frac{\text{g}}{\text{cm}^3}$$

$$25 \text{ gallons of White Oil} = 75,708 \text{ grams of White Oil}$$

$$0.5 \text{ wt\% Acetic Acid} = (0.05) * (75,708) = 378.41 \text{ grams}$$

$$\text{Initial Acetic Acid added} = \frac{\text{grams Acetic acid}}{\text{Acetic acid Density}}$$

$$\text{Initial Acetic Acid added} = 360.5 \text{ mL}$$

Sample Calculations for Extract:

Moles of NaOH Givens:

$$\text{NaOH Delivered} = .0618 \text{ L}$$

$$\text{Normality of NaOH} = 0.01$$

$$\text{Moles of NaOH} = (\text{NaOH Delivered} * \text{Normality of NaOH})$$

$$\text{Moles of NaOH} = .00618 \text{ moles}$$

Due to Normality given of NaOH, the acetic acid moles will equal the moles of NaOH.

Mass of Acetic Acid Givens:

$$\text{Moles of Acetic acid} = .00618 \text{ moles}$$

$$\text{Molecular Weight of Acetic acid} = 60.052 \frac{\text{g}}{\text{mol}}$$

$$\text{Mass of Acetic acid} = (\text{Moles of Acetic acid} * \text{Molecular Weight of Acetic acid})$$

$$\text{Mass of Acetic acid} = .0371 \text{ grams}$$

Mass of Water Givens:

Sample Volume= 25 mL

Mass of acetic acid= .0371 grams

Density of Acetic acid= 1.05

Density of Water= .997

$$\text{Mass of Water} = (\text{Sample Volume} - \left(\frac{\text{Mass of Acetic acid}}{\text{Density of Acetic acid}} \right) * \text{Density of Water})$$

$$\text{Mass of Water} = 24.89 \text{ grams}$$

Moles of Water Givens:

Mass of Water= 24.89 grams

Molecular Weight of Water= 18.02 $\frac{\text{g}}{\text{mol}}$

$$\text{Moles of Water} = 1.382 \text{ moles}$$

Mole fraction of Acetic Acid Givens:

Moles of Acetic Acid= .00618 moles

Moles of Water= 1.382 moles

$$\text{Mole Fraction of Acetic Acid} = (\text{Moles of Acetic acid} / (\text{Moles of Acetic acid} + \text{Moles of Water}))$$

$$\text{Mole Fraction of Acetic Acid} = .000447$$

Mass fraction of Acetic Acid Givens:

Mass of Acetic Acid= .0371 grams

Mass of Water= 24.89 grams

$$\text{Mass Fraction of Acetic Acid} = (\text{Mass of Acetic acid} / (\text{Mass of Acetic acid} + \text{Mass of Water}))$$

$$\text{Mass Fraction of Acetic Acid} = .00162$$

Weight Fraction of Acetic Acid Given:

Mass Fraction of Acetic acid=.00162

*Weight Fraction of Acetic Acid = (Mass fraction of Acetic acid * 100)*

Weight Fraction of Acetic Acid = .162

-The same equations were done above for the raffinate.

Murphree Efficiencies for raffinate and extract of 400 RPM Stage 1

$$E_{ME} = \left(\frac{0.057 - 0.149}{0.07 - 0.149} \right) = 1.17$$

$$E_{MR} = \left(\frac{0.269 - 0.078}{0.269 - 2.387} \right) = 0.09$$

Overall Efficiency for 400 RPM

$$E_o = \left(\frac{1}{3} \right) = 0.33$$

10. Appendix IV: Individual Team Contributions

NAME: Zainab

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Transferred Samples from apparatus to titration setup
Pre-Lab Editing	0.5	Apparatus and Materials
Experimental Objective (Prelab)	0	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	0.5	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	1	Cohesiveness and References
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	1	Calculations, Tables, and Graphs
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	1	Made tables and Graphs
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	1	Wrote sample calculations
Power Point Presentation	0	

Total	14	
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NAME: Ariel

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8 hours	
Pre-Lab Editing	15min	
Experimental Objective (Prelab)	0min	
Apparatus (prelab)	0min	
Materials & Supplies (prelab)	0min	
Procedure (prelab)	30min	
Project Safety Evaluation (prelab)	30min	
Final Lab Editing	30 min	
Executive Summary (final lab)	1 hr	
Introduction (Final lab)	1 hr	
Theory (Final Lab)	0min	
Results (final lab)	0min	
Discussion (final lab)	0min	
Conclusion (final lab)	0min	
References (final and/or prelab)	0min	
Data Tabulation/Graphs (final)	0min	
Error Analysis (final lab)	1hr	Editing
Sample Calculations (final Lab)	15min	Editing
Power Point Presentation	1 hr	Created Powerpoint

Total	14	
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NAME: Tom

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8 hours	
Pre-Lab Editing	10 min	
Experimental Objective (Prelab)	40 min	
Apparatus (prelab)	1 hr	
Materials & Supplies (prelab)	40 min	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	2 hours and 30 minutes	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	30 min	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	

Total	13.5	
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NAME: Khalil

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8 hours	
Pre-Lab Editing	10 minutes	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	30 min	Did procedure
Project Safety Evaluation (prelab)	30 min	Did project safety
Final Lab Editing	1 hour	Edit of final lab
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	2 hours	Wrote discussion for results
Conclusion (final lab)	30 min	Conclusion made based off results
References (final and/or prelab)	30 min	Gathered references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	30 min	Wrote out errors
Sample Calculations (final Lab)	0	

Power Point Presentation	0	
Total	13 hours and 40 minutes	